THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The power of a machine that does the same work in less time is greater. Reason (R): Power = work/time. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The work done by the net force is equal to: 1

(a) the change in KE **(b)** the change in speed
(c) the change in PE only **(d)** the weight

3. A particle of mass 1 kg moves in a horizontal circle at 5 m/s. The work done by the centripetal force in a half revolution is: 1

(a) 25 J **(b)** 50 J **(c)** 12.5 J **(d)** zero

4. A machine lifts a load of 400 N with an effort of 100 N. Its mechanical advantage is: 1

(a) 4 **(b)** 500 **(c)** 1/4 **(d)** 300

5. Assertion (A): Work done by the centripetal force in uniform circular motion is zero. Reason (R): The force is perpendicular to the displacement at every instant. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A 1 kg ball dropped from 10 m loses 20 per cent of its energy on rebound. It rises to (g = 10): 1

(a) 5 m **(b)** 10 m **(c)** 8 m **(d)** 2 m

7. A body at rest is acted on by a force of 20 N. After moving 5 m its KE is 60 J. The friction force is: 1

(a) 12 N **(b)** 8 N **(c)** 20 N **(d)** 4 N

8. A lever of length 2 m has a load of 300 N at one end. The fulcrum is 40 cm from this end. The effort at the other end is: 1

(a) 75 N **(b)** 300 N **(c)** 187.5 N **(d)** 120 N

9. **Assertion (A):** At the highest point of a ball thrown upward, its potential energy is maximum. 1
Reason (R): At that point its kinetic energy is zero and its mechanical energy is entirely potential (ignoring air resistance).
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The total mechanical energy of a freely falling body (ignoring air resistance) remains constant. 1

Reason (R): The work done by gravity on a falling body is positive.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain the energy conversions in a rubber ball bouncing. 2

12. A force of 12 N moves a block of mass 3 kg on a smooth surface by 5 m. Find the final speed from rest. 2

13. A body of mass 5 kg moves from rest under a force that does 250 J. Find the speed and momentum. 2

14. What is the minimum work needed to lift a 20 kg bag onto a shelf 1.5 m high and what is the power if done in 3 s (g = 10)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A weightlifter lifts a 100 kg barbell to 2 m in 2 s and holds for 5 s. Find the work and power in the lift and the work done during the hold (g = 10). 3

16. A motorcyclist moving at 36 km/h accelerates to 72 km/h in 5 s. If the mass of the bike and rider is 200 kg, find the work done and the average power. 3

17. Why do a hammer and a nail make a better combination than pushing the nail by hand? Use the force-distance relation of work. 3

OR

A cricket ball of mass 0.16 kg is hit and flies at 25 m/s. Find the KE and the height it would reach if thrown vertically upward (g = 10). 3

18. Calculate the power of a waterfall 20 m high with a flow rate of 500 kg per second if 40 per cent is converted to electricity (g = 10). 3

OR

A person lifts a 15 kg bag to the shoulder of height 1.5 m using a constant force. If he then walks 10 m horizontally, find the work done against gravity during the walk and in the lift (g = 10). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A mechanic uses a crowbar as a lever to lift a load of 600 N. The effort arm is 1.5 m and the load arm is 0.5 m. The effort end is pushed down through 0.6 m. 5

(i) Find the mechanical advantage of the lever.

(ii) Find the effort required, assuming an ideal lever.

(iii) How high does the load rise?

(iv) Find the work done by the effort and the work done on the load, and say why the efficiency of a real machine is below 100 per cent.

20. A body of mass 5 kg is thrown vertically up with 30 m/s. Plot (in words) the KE, PE and the total energy as functions of height; find the height where KE = PE and the time to reach it (g = 10). 5

OR

Explain the conservation of energy in a system with friction: a block slides from a height of 2 m down an incline and stops after sliding 5 m on a rough floor. Find the friction force and the heat produced for a mass of 10 kg (g = 10). 5

END OF QUESTION PAPER — Total: 40 Marks